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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Course : Broad Discipline of COMP-61431, BEng Electrical Engineering-41470

Subject : COMP4423 Computer Vision

Group : 201, 2011

Session : 2022 / 2023 Semester II

Date : 22 April 2023 Time : 12:30 - 14:30

Time Allowed: 2 Hours Subject Lecturer: Prof. WEI Xiaoyong

This question paper has 6 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

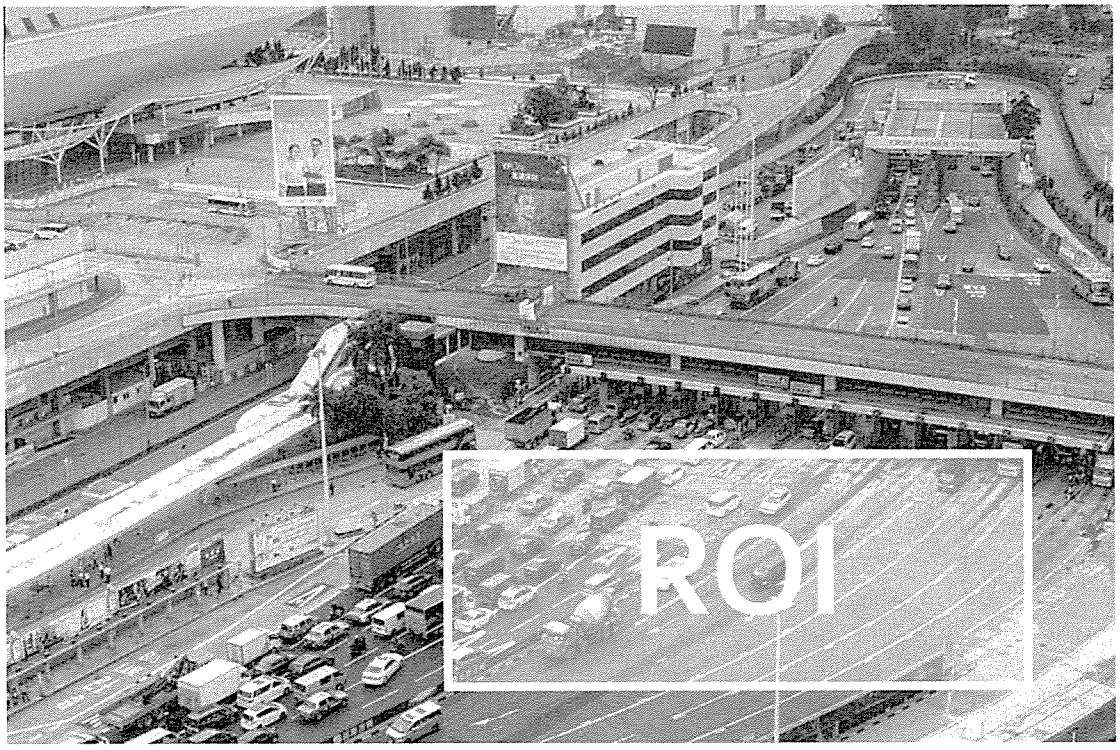
The exam is an open-book exam. Students are allowed to bring 5 pages of A4 size material to the exam. Electronic devices are not allowed. Students should write their answers on the answer sheet. Only the answers on the answer sheet will be graded.

Do not turn this page until you are told to do so!

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Section D: Short-Answer Questions (30 marks)

25. You are invited as a computer vision expert to assist the Cross-Harbor Tunnel to develop a better strategy for management. They want you to create a solution to analyze the dynamic traffic load of the tunnel. You propose installing an oblique view camera which captures images of the traffic every minute (see the sample image below), on the basis of which you can use what you have learned in COMP 4423 to help them out.



You decide to start from the simplest case by estimating the number of vehicles in a region of interest (ROI, shown as a rectangle in the sample image). To explain your approach, please answer each of the following questions with a short answer in 10 words unless specified. You can assume the camera is well fixed so that exactly the same ROI will be captured in images over time t, t+1, t+2, … (i.e., a background point appears at exactly the same pixel location (x,y)). You can also assume: a) only vehicles enter the ROI, with no other objects (e.g., birds, pedestrians); and b) Objects in the ROI have no shadows.

a. Estimate the number of vehicles per ROI using pure image processing methods (i.e., without any machine learning techniques)
    i. What are the necessary steps to take? (20 words, 5 points)
    ii. Identify three main factors influencing performance? (3 points)
b. Now you are allowed to use machine learning to create the solution
    i. Name a pretrained model you can use for this task. (2 points)
    ii. Name the primary challenge by adopting the pretrained model and what is the technique to address it. (2 points)
    iii. Explain if the image processing-based method you built in part-a helps to reduce the effort in part b-ii and how? (3 points)
c. Now you decided to move forward by classifying the vehicles into 4 categories: cars,

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trucks, motorcycles, and others. You can assume the vehicles are well separated by the methods you built above.

i. Design a simple CNN model with up to 6 layers and draw a figure to illustrate its architecture. Please label the type of each layer and necessary components. Let us assume the input layer is with a shape of 64x64x3. (30 words, 10 points)

ii. The model is trained in April using data collected from January to March. It performs well until May. What are the most possible reasons? Please identify two of them. (2 points)

iii. Please suggest solutions to address these issues. (20 words, 3 points)

** END **